

BlinkIT Grocery Outlet Sales Dashboard

Combined System Design Document (SDD) & Project Design Document (PDD)

Prepared by: Madhumitha **Platform:** Microsoft Power BI Desktop

System Design Document (SDD)

BlinkIT Grocery Outlet Sales Dashboard — Power BI

Project	BlinkIT Grocery Outlet Sales Dashboard
Prepared by	Madhumitha
Platform	Microsoft Power BI Desktop
Document type	System Design Document

1. Introduction

1.1 Purpose

This document describes the technical design of the BlinkIT dashboard — its architecture, data model and report layout — so the system can be understood, maintained and extended.

1.2 Scope

Covers the Power BI `.pbix` file only: the sales/outlet table, the field-parameter used for dynamic switching, the measures, and the single-page report built on top of them.

1.3 Definitions

Term	Meaning
SDD	System Design Document (this document)
PDD	Project Design Document (companion document)
Field parameter	A Power BI table that lets a visual switch between measures dynamically

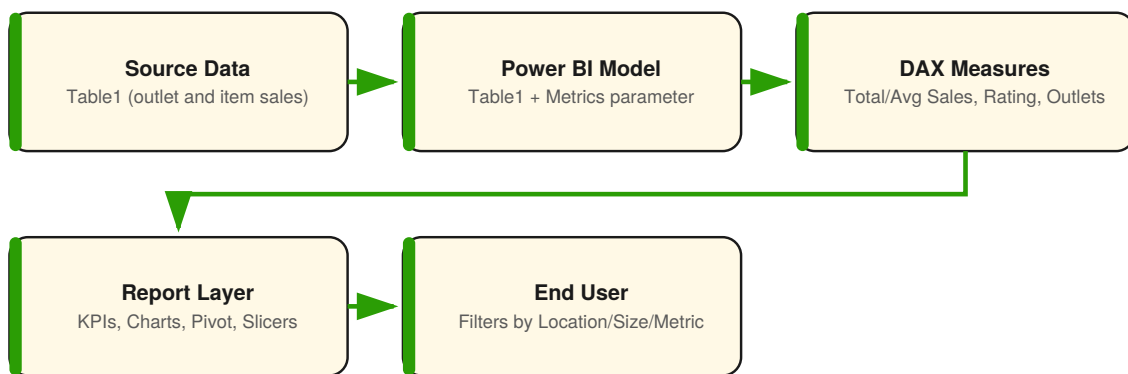
2. System Overview

The dashboard gives a retail/category team a single-page view of grocery outlet performance — total and average sales, average customer rating, outlet count, and how sales vary by outlet size, location type, outlet type, item type and item fat content.

3. System Architecture

Layers

- Source data** — outlet and item-level sales records (`Table1`).
- Data model** — the fact/dimension columns loaded into a single table, plus a disconnected `Metrics` field-parameter table for dynamic measure switching.
- Measure layer** — four core DAX measures computed once and reused across visuals.
- Report layer** — KPI cards, charts, a pivot table and slicers on one page.
- Consumption** — the end user filters by Outlet Location Type, Outlet Size, or switches the active Metric; nothing is written back to the source.



4. Data Model

4.1 Tables

Table	Role
Table1	Main fact table — one row per item/outlet sales record
Metrics	Disconnected field-parameter table used to switch the active measure in a chart

4.2 Data Dictionary — key columns (Table1)

Column	Used for
Item Fat Content	Clustered bar chart (sales by fat content)
Item Type	Funnel chart, pivot table
Outlet Establishment Year	Area chart (sales trend by year opened), pivot table
Outlet Location Type	Funnel chart, slicer
Outlet Size	Donut chart, pivot table, slicer
Outlet Type	Pivot table

5. Measures

Measure	Purpose
Total sales	Sum of sales value across the current filter context
Average sales	Average sales value per record/outlet
Average Rating	Average outlet/item rating
Total Outlets	Count of distinct outlets

All four are used on the KPI cards and reused across the charts and pivot table, so every number on the page traces back to the same four definitions.

6. Report / UI Design

Visual	Type	Fields	Purpose
Brand header	Text box + shapes	“BLINKIT” wordmark	Page branding
KPI cards	Card visual	Total sales, Average sales, Average Rating, Total Outlets	Headline performance at a glance
Outlet Location Type slicer	Slicer	Outlet Location Type	Filter by Tier 1/2/3 location
Outlet Size slicer	Slicer	Outlet Size	Filter by Small/Medium/High

			outlet size
Metrics slicer	Slicer	Metrics (field parameter)	Switch the active measure shown in a chart
Outlet Size split	Donut chart	Outlet Size, Average sales	Share of average sales by outlet size
Sales by location	Funnel chart	Outlet Location Type, Average sales	Compare average sales across location tiers
Sales trend by year	Area chart	Outlet Establishment Year, Average sales	Trend of average sales by outlet age
Fat content comparison	Clustered bar chart	Item Fat Content, Average sales, Total sales	Compare low-fat vs. regular item sales
Sales by item type	Funnel chart	Item Type, Total sales	Rank item categories by total sales
Detail breakdown	Pivot table	Item Type, Outlet Establishment Year, Outlet Size, Outlet Type, Average Rating, Average sales	Drill into any combination of dimensions
Decorative shapes	Shape	—	Visual layout/branding accents

All visuals share the same filter context from `Table1` , so the slicers and the Metrics parameter update every relevant visual together.

7. Non-Functional Requirements

- **Usability** — single page, clearly branded, three slicers, no nested navigation.
- **Performance** — a single flat fact table with pre-defined measures keeps visuals responsive.
- **Maintainability** — centralized measures and a field-parameter table make it easy to add a new metric without duplicating chart logic.
- **Portability** — self-contained `.pbix` ; no external services required to open it.

8. Tools & Technology Stack

- **Power BI Desktop** — data modeling, DAX, report design
- **DAX** — measure calculations
- **Power BI field parameters** — dynamic metric switching

9. Assumptions & Constraints

- Source sales/outlet data is assumed clean and pre-loaded into `Table1` ; this document covers the model and report layer only.
- The report is a single page; it is not built for drill-through or multi-page navigation.

Project Design Document (PDD)

BlinkIT Grocery Outlet Sales Dashboard — Power BI

Project	BlinkIT Grocery Outlet Sales Dashboard
Prepared by	Madhumitha
Document type	Project Design Document

1. Project Overview

1.1 Problem Statement

Grocery outlet sales data spans many outlets, item categories and locations, making it hard to see at a glance which outlet types, sizes and item categories actually drive sales.

1.2 Objective

Deliver a single Power BI dashboard that shows total and average sales, ratings and outlet count, broken down by outlet size, location type, outlet type, item type and item fat content — with the ability to switch the active metric on demand.

1.3 Scope

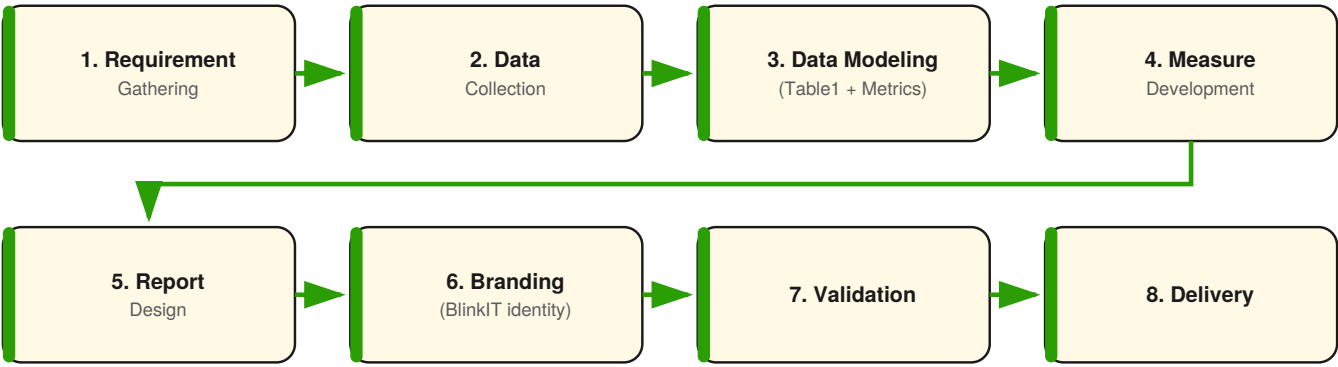
In scope: sales and rating KPIs, outlet-size and location-type breakdowns, item-type and fat-content comparison, a drillable pivot table, dynamic metric switching. **Out of scope:** inventory/stock management, price optimization, forecasting, multi-page drill-through, automated data refresh scheduling.

2. Stakeholders

Stakeholder	Interest
Category/Retail Managers	Which item types and outlet formats drive sales
Operations Team	Performance by outlet size and location tier
Business Analysts	Ad-hoc exploration via the pivot table and slicers
Project Owner (Madhumitha)	Build, validate and maintain the dashboard

3. Project Methodology / Process Flow

Step	Description
1. Requirement gathering	Define which sales questions the dashboard needs to answer
2. Data collection	Source outlet and item-level sales records
3. Data modeling	Load into <code>Table1</code> ; add the <code>Metrics</code> field-parameter table
4. Measure development	Build Total sales, Average sales, Average Rating, Total Outlets
5. Report design	Lay out KPI cards, donut, funnel, area, bar charts, pivot table, slicers
6. Branding	Apply the BlinkIT wordmark and yellow/green visual identity
7. Validation	Cross-check visual totals against the source data
8. Delivery	Publish the final <code>.pbix</code> and supporting docs



4. Deliverables

Deliverable	Description
BlinkIT_Dashboard_powerbi.pbix	The Power BI report file
SDD.md	System Design Document (architecture, data model, measures, visuals)
PDD.md	This Project Design Document (process, scope, roles)

5. Roles & Responsibilities

Role	Responsibility	Owner
Data Analyst / Developer	Data modeling, DAX, report build, branding	Madhumitha
Reviewer	Validate figures against source data	Madhumitha

6. Risks & Mitigation

Risk	Mitigation
Outlet/item categories inconsistent (e.g. “Low Fat” vs “LF”)	Standardize category labels during data load
Field-parameter (Metrics) confuses first-time viewers	Label the slicer clearly and default to the most-used metric
Pivot table becomes dense as categories grow	Keep slicers as the primary way to narrow focus before reading the pivot

7. Success Criteria

- Total sales, average sales, average rating and outlet count are visible without scrolling.
- Outlet Location Type, Outlet Size and Metrics selections correctly update every relevant visual.
- KPI and chart figures reconcile with the source sales data.

8. Future Enhancements

- Add a time-trend view beyond outlet establishment year (e.g. monthly/seasonal sales).
- Outlet-level drill-through page.
- Forecasted sales alongside actuals.